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Subject :- Data Analytics

Division :- SA2

Github Link :- https://github.com/Himanshu-git1414/Data-Analytics-Assignment

Dataset Description :- Rows :- 1500 Columns :- 5

Title :- Analysis of Student Attendance and Study Habits on Academic Performance.

Problem Statement :- I am analyzing student attendance and study habits to understand how they influence academic performance. This study aims to identify whether low attendance and inconsistent study patterns lead to poorer grades, so that institutions can support students more effectively.

In [8]:
```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
```

In [9]:
```
df = pd.read_csv("student_performance_dataset Himanshu.csv")
df.head()
```

Out[9]:

	Student_ID	Study_Hours_Weekly	Attendance_Rate_Percent	Final_Grade	Attendence_Cate
0	1001	18.5	92.8	55.3	
1	1002	14.0	79.5	53.2	Av
2	1003	19.5	76.8	59.7	Av
3	1004	25.7	85.0	78.8	Av
4	1005	13.4	83.3	60.6	Av

In [10]:
```
df.shape
```

Out[10]:
```
(1500, 5)
```

I observed that my dataset contains 1500 rows & 5 columns

In [11]:
```
df.dtypes
```

Out[11]:
```
Student_ID                   int64
Study_Hours_Weekly         float64
Attendance_Rate_Percent    float64
Final_Grade                float64
Attendence_Categroy         object
dtype: object
```

In my dataset Col1 have integer data, Col2 & Col3 & Col4 have float variables & last column contains string data

In [12]:
```
df.isnull().sum()
```

Out[12]:
```
Student_ID                 0
Study_Hours_Weekly         0
Attendance_Rate_Percent    0
Final_Grade                0
Attendence_Categroy        0
dtype: int64
```

No missing values Found

In [13]:
```
df.drop_duplicates(inplace=True)
df.shape
```

Out[13]:
```
(1500, 5)
```

No duplicates value

In [14]:
```
Mean = df["Study_Hours_Weekly"].mean()
Median = df["Study_Hours_Weekly"].median()
Mode = df["Study_Hours_Weekly"].mode()
Standard_Deviation = df["Study_Hours_Weekly"].std()
Variable = df["Study_Hours_Weekly"].var()
Range = df["Study_Hours_Weekly"].max()-df["Study_Hours_Weekly"].min()
Mid_range = (df["Study_Hours_Weekly"].max()-df["Study_Hours_Weekly"].min()) / 2
print("Statistics of Study Hours :- ", Mean, Median, Mode, Standard_Deviation, Vari
```

```
Statistics of Study Hours :-  15.371733333333333 15.350000000000001 0    0.0
Name: Study_Hours_Weekly, dtype: float64 6.853700445897113 46.97320980209029 40.0 2
0.0
```

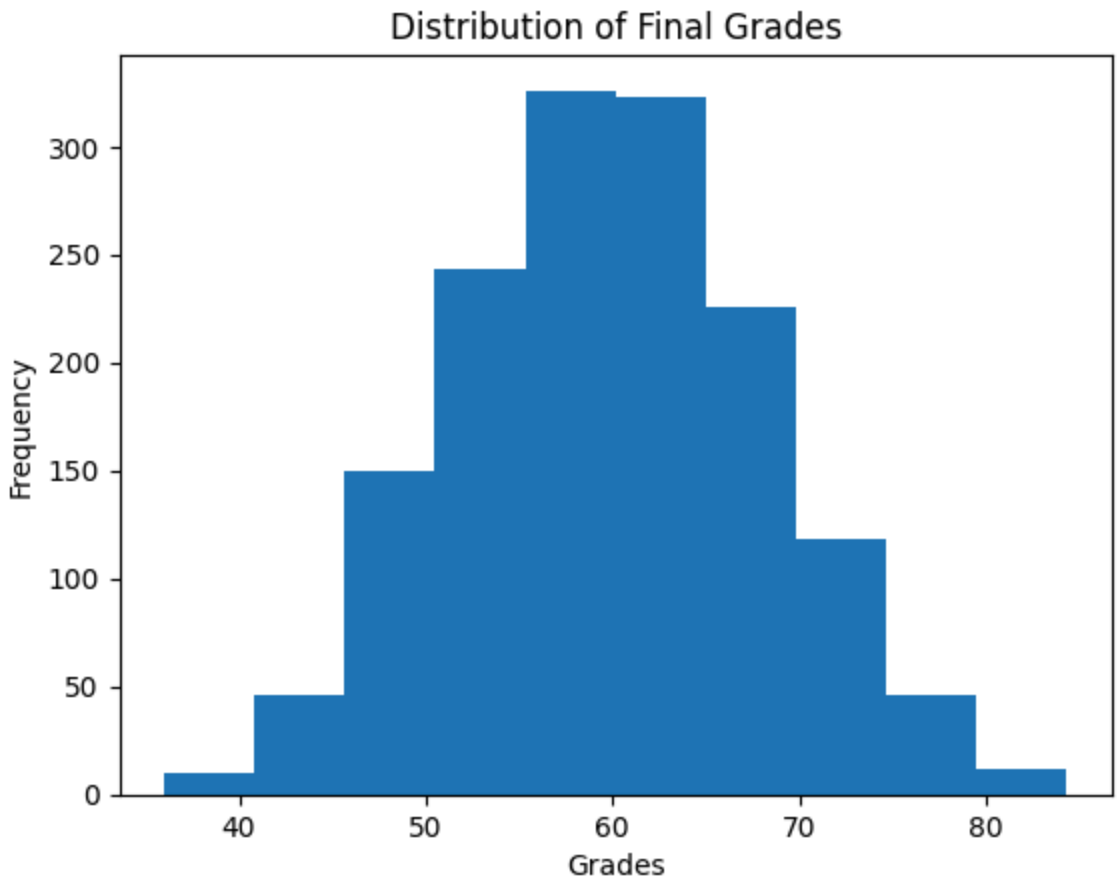
In [15]:
```
Meanfg = df["Final_Grade"].mean()
Medianfg = df["Final_Grade"].median()
Modefg = df["Final_Grade"].mode()
Standard_Deviationfg = df["Final_Grade"].std()
Variable = df["Final_Grade"].var()
Range = df["Final_Grade"].max()-df["Final_Grade"].min()
Mid_range = (df["Final_Grade"].max()-df["Final_Grade"].min()) / 2
print("Statistics of Final Grade :- ", Meanfg, Medianfg, Modefg, Standard_Deviation
```

```
Statistics of Final Grade :-  59.837666666666664 59.8 0    54.6
Name: Final_Grade, dtype: float64 8.243454630901015 67.95454425172338 48.3 24.15
```

Visualizations

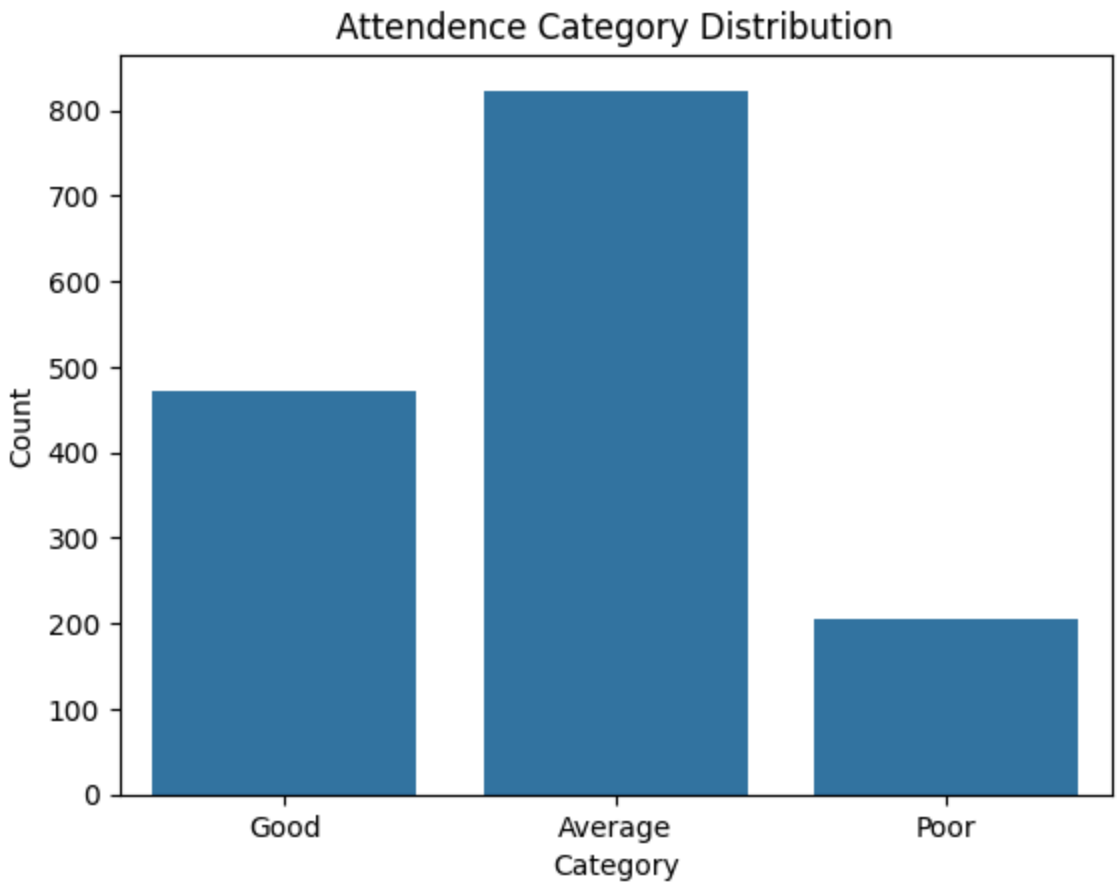
Histogram

In [16]:
```
plt.hist(df["Final_Grade"])
plt.title("Distribution of Final Grades")
plt.xlabel("Grades")
plt.ylabel("Frequency")
plt.show()
```



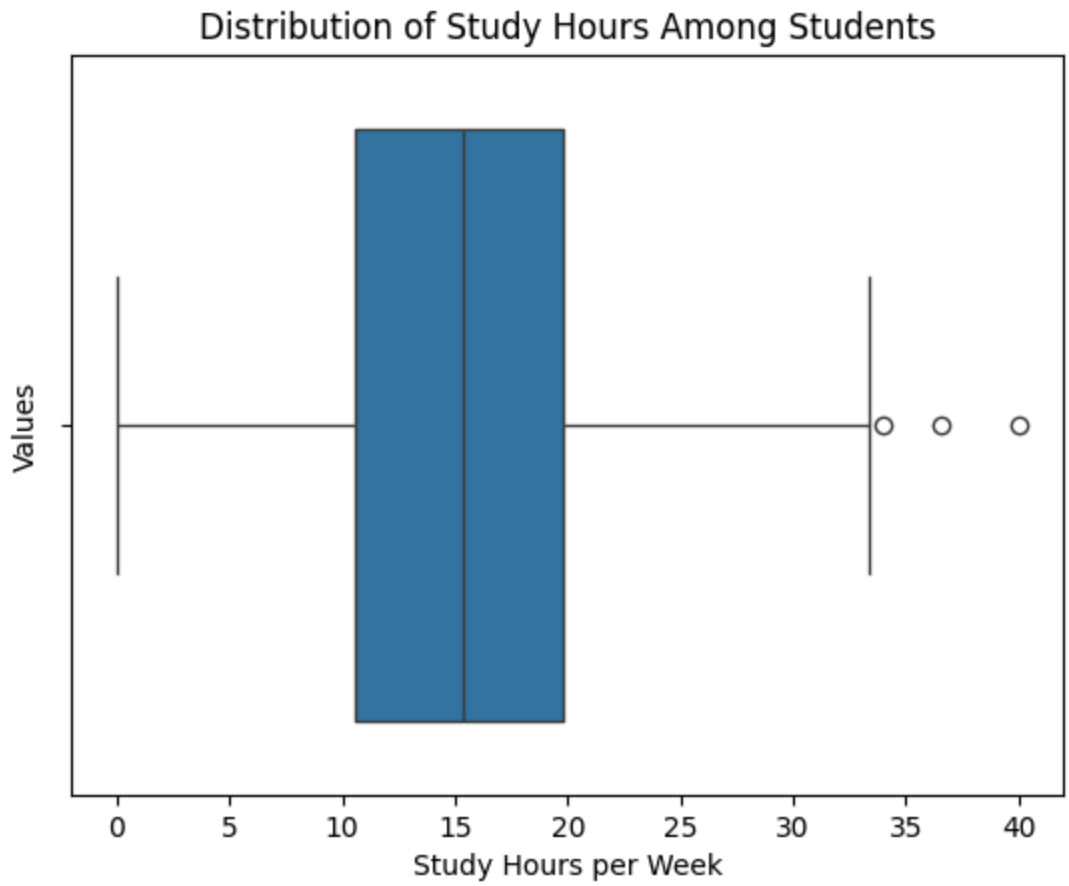
Bar Chart

In [17]:
```
sns.countplot(x = "Attendence_Categroy", data = df)
plt.title("Attendence Category Distribution")
plt.xlabel("Category")
plt.ylabel("Count")
plt.show()
```



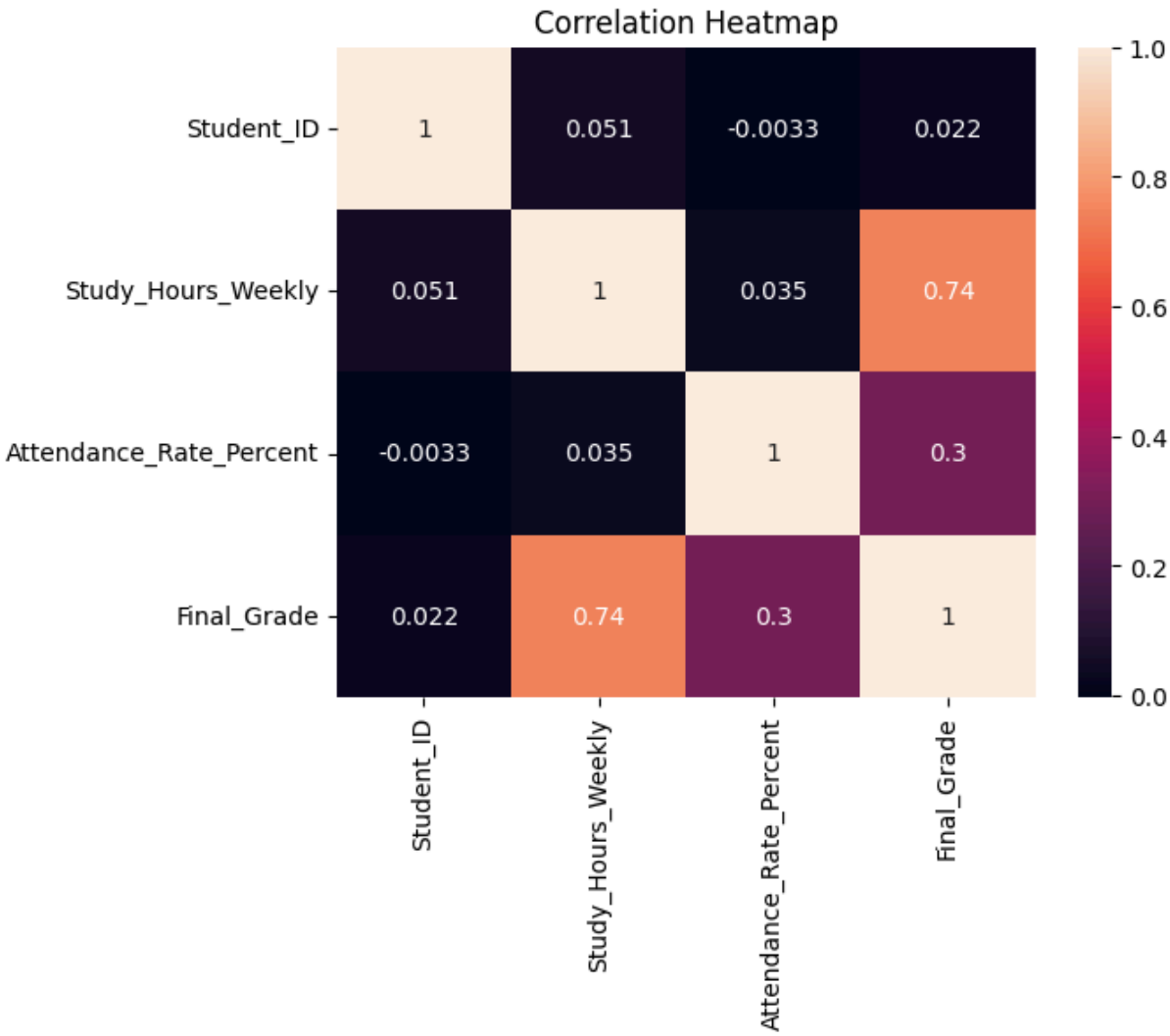
Boxplot

In [18]:
```
sns.boxplot(x = df["Study_Hours_Weekly"])
plt.title("Distribution of Study Hours Among Students")
plt.xlabel("Study Hours per Week")
plt.ylabel("Values")
plt.show()
```



Heatmap

In [19]:
```
sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.title ("Correlation Heatmap")
plt.show()
```



Most students have grades between 50-70.

Encode categorical

In [20]:
```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()
df['Attendence_Categroy'] = le.fit_transform(df['Attendence_Categroy'])
df
```

Out[20]:

	Student_ID	Study_Hours_Weekly	Attendance_Rate_Percent	Final_Grade	Attendence_
0	1001	18.5	92.8	55.3	
1	1002	14.0	79.5	53.2	
2	1003	19.5	76.8	59.7	
3	1004	25.7	85.0	78.8	
4	1005	13.4	83.3	60.6	
...	...	...	...	...	
1495	2496	29.0	85.8	81.0	
1496	2497	29.4	87.6	69.4	
1497	2498	23.5	72.6	69.6	
1498	2499	22.2	88.3	56.2	
1499	2500	19.1	83.4	57.1	

1500 rows × 5 columns

Train Test split

In [21]:

```
from sklearn.model_selection import train_test_split

X = df[['Study_Hours_Weekly', 'Attendance_Rate_Percent']]
y = df['Final_Grade']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
print(X_train)
print(X_test)
print(y_train)
print(y_test)
```

```
      Study_Hours_Weekly  Attendance_Rate_Percent
1250                17.3                    100.0
515                 20.3                     83.1
1473                15.1                    100.0
293                 20.9                     91.7
45                  10.0                     89.7
...                  ...                      ...
608                  2.5                     81.7
550                 15.6                     93.3
1342                18.1                     99.8
987                 21.1                     84.5
1348                 5.4                     78.9

[1200 rows x 2 columns]
      Study_Hours_Weekly  Attendance_Rate_Percent
514                 11.0                     93.3
966                 10.3                     77.4
1227                14.0                     98.4
1027                15.8                     91.1
833                  9.6                     81.4
...                  ...                      ...
860                 16.4                     74.7
221                  1.9                     82.4
1321                18.7                     93.5
1194                 3.8                     88.3
479                 22.8                     89.1

[300 rows x 2 columns]
1250    71.0
515     66.4
1473    66.3
293     68.5
45      46.1
        ...
608     41.0
550     59.7
1342    75.9
987     62.4
1348    44.9
Name: Final_Grade, Length: 1200, dtype: float64
514     60.4
966     55.1
1227    64.3
1027    60.7
833     46.9
        ...
860     56.2
221     51.1
1321    64.7
1194    38.6
479     74.8
Name: Final_Grade, Length: 300, dtype: float64
```

Model

In [22]:
```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)
predictions = model.predict(X_test)
print(predictions)
```

```
[58.12098686 53.62795933 61.97244163 61.75187227 53.99745114 60.39573434
 47.55261222 62.79035819 62.49185686 53.33474208 47.81709526 67.6984486
 63.98139323 62.61468714 72.70726468 55.12706287 62.19136993 52.9625919
 56.20880604 60.18243467 50.95367298 66.32350452 71.9692657  70.15118095
 49.15572356 56.85005058 54.36364448 68.81783673 66.9792721  51.74416817
 55.84228448 60.5542403  58.75266421 64.33833127 56.34419008 66.95352448
 65.96853575 57.74754016 45.46210199 62.21019241 61.42069818 66.38888272
 69.84938116 53.60418099 55.44108821 66.3581629  55.50183876 59.41074562
 57.09407007 64.78079918 70.91791414 58.7463955  70.0616962  58.0810444
 60.97657288 61.72612465 68.44734394 54.40393148 50.78064397 65.92396565
 49.54999469 49.41889375 65.92394932 57.3450147  55.88652637 67.15559957
 50.37614924 70.87398413 60.24846929 61.15949728 63.67000992 57.34468649
 68.18912962 46.13009512 60.6447097  70.58242428 66.86403972 57.54577694
 65.86682545 61.32855506 66.7157737  59.62471805 72.17662487 64.36507987
 60.18937614 53.38064137 52.73476919 58.16952818 62.01174399 57.32025172
 53.43115198 53.60979328 58.7985635  57.36219612 62.10286981 65.54093544
 55.19508311 53.15378696 59.69935155 59.86908209 57.89314781 58.14410877
 70.40015631 62.92474126 74.30674934 62.27260035 67.58155882 62.66884076
 62.47104877 58.38647087 71.49245129 53.24855579 63.62873829 55.66661344
 58.61101145 68.38626518 66.11547259 73.82663647 74.62374493 58.30624141
 50.517802   56.63177871 58.40297952 51.35089802 53.12605373 71.85831642
 58.02324777 59.57783412 60.96764579 61.20605299 54.83615945 58.54860351
 57.54809077 67.13249398 61.79512952 52.7486358  49.60710223 65.38177305
 59.01022211 64.31059804 72.08978215 54.42208121 58.8949734  51.56122744
 61.00496254 60.43800695 60.01436153 63.0350341  57.85386178 63.64756077
 44.53655096 57.8274414  54.50728286 60.06521668 56.78004473 57.89976107
 59.18622137 59.15187487 68.18185993 54.16816632 62.41459766 56.61891306
 58.43534041 60.29701062 63.53231206 65.75093664 49.33469307 70.79936696
 62.41789613 56.02387969 55.30900263 43.65689922 58.91313945 48.10007257
 54.3253431  63.10305434 60.72429906 63.10139694 54.46071081 70.46850475
 53.08709591 68.72538172 54.00701832 48.85887964 62.93201095 53.08808055
 51.60745494 59.37971392 45.09855068 56.41485236 64.63089209 61.52634705
 47.22107725 60.13059488 66.44170715 61.43653408 46.48242185 51.10159445
 58.06189371 67.15592778 56.92071285 53.39285059 66.91323748 61.1542132
 51.55065929 62.00414609 61.01784452 55.70128815 67.32004603 65.23153775
 56.41056925 62.3752953  60.69722225 66.63323042 58.41684614 46.32096197
 53.21156726 56.96892596 66.48297879 50.93847717 54.83880149 61.67791153
 62.24189687 58.15764718 65.75290592 57.32652043 56.33560754 62.16230751
 62.62228504 50.20541772 56.40065753 51.52061223 58.36863303 65.0750174
 50.20080641 65.68752773 55.92847077 66.13396686 59.98398625 54.23717119
 63.60297433 56.58192453 55.58769684 61.15982549 47.3505208  53.47870867
 76.44315886 40.43150951 62.54369666 63.82817134 51.62859125 70.91459934
 48.72085356 63.47287436 61.50918197 59.96152076 49.48361552 56.7853288
 56.8774556  66.01906269 72.9998255  55.14688633 47.92175949 56.85764848
 59.35165247 53.4559313  68.89642511 60.36007499 64.24159316 59.61119598
 65.65352577 67.29098361 57.04321491 53.3317555  66.35916387 56.37125056
 66.72601364 56.35078701 51.70420939 63.36655274 64.81910057 62.41458133
 59.51642715 51.9739765  63.79845249 56.22564291 53.05901813 65.48973574
 66.50939917 71.12791535 55.43746154 56.8893366  56.44621227 50.74662569
 62.11872204 58.2656262  47.55490971 64.85674553 50.64657278 67.34216698]
```

A Linear Regression model was used to predict students' final grades based on study hours and attendance rate. The model was trained on 80% of the data and tested on the remaining 20%.

Findings

Finding 1:- The histogram shows that most students have final grades in the mid-range, indicating that the majority of students perform at an average level, with fewer students achieving very high or very low scores.

Finding 2:- The countplot shows that most students fall into the ‘Average’ attendance category, while fewer students have ‘Good’ or ‘Poor’ attendance, indicating moderate lecture participation among students.

Finding 3:- The boxplot shows a wide spread in study hours, with some students studying significantly more or less than others. This indicates variation in study habits, and the presence of lower values suggests that some students may rely on last-minute studying.

Finding 4:- he heatmap shows a positive correlation between attendance rate, study hours, and final grades, indicating that students who attend lectures regularly and study consistently tend to achieve higher academic performance.

Recommendations

1. Colleges should monitor student attendance regularly and identify students with low attendance early. Providing academic support, mentoring, or reminders can help improve their participation and overall performance.

2. Students should be encouraged to follow consistent study habits instead of last-minute preparation. Workshops on time management and study planning can help students improve their learning outcomes.